

1. $\int_{0^-}^{\infty} \delta(t) e^{-st} dt = e^{-s \cdot 0} = 1 //$ (sampling property of the impulse) ①

1a. $\int_{0^-}^{\infty} \underbrace{\delta(t-T)}_{\substack{\uparrow \\ \text{zero when } t \neq T}} e^{-st} dt = e^{-s \cdot T} //$ (sampling property of the impulse)

2. $\int_{0^-}^{\infty} \underbrace{u(t)}_{\substack{\uparrow \\ =1 \text{ when } t > 0}} e^{-st} dt = \int_{0^+}^{\infty} 1 \cdot e^{-st} dt$

$$= \left. \frac{e^{-st}}{(-s)} \right|_{t=0^+}^{t \rightarrow \infty}$$

$$= \frac{e^{-\infty}}{(-s)} - \frac{e^0}{(-s)} = 0 + \frac{1}{s}$$

$$= 1/s //$$

2a. $\int_{0^-}^{\infty} u(t-T) e^{-st} dt$

Applying the time shift property (#4) to result 2 above tells us that the answer is $e^{-sT} \cdot \frac{1}{s}$

Let's try to do it from scratch:

Let $\tau = t - T \Rightarrow d\tau = dt$ (change of variable)

$$\Rightarrow \int_{-T}^{\infty} u(\tau) e^{-s(\tau+T)} d\tau = e^{-sT} \int_{-T}^{\infty} \underbrace{u(\tau)}_{\substack{\uparrow \\ =1 \text{ when } \tau > 0}} e^{-s\tau} d\tau$$

$$= e^{-sT} \int_{0^+}^{\infty} 1 \cdot e^{-s\tau} d\tau = e^{-sT} \left[\frac{e^{-s\tau}}{(-s)} \right]_{0^+}^{\infty} = \frac{e^{-sT}}{s} //$$

(2)

$$\begin{aligned}
 3. \int_{0^-}^{\infty} (e^{-at} u(t)) e^{-st} dt &= \int_{0^-}^{\infty} \underbrace{u(t)}_{=1 \text{ when } t > 0} e^{-(s+a)t} dt \\
 &= \int_{0^+}^{\infty} e^{-(s+a)t} dt = \left. \frac{e^{-(s+a)t}}{-(s+a)} \right|_{t=0^+}^{t \rightarrow \infty} \\
 &= \frac{e^{-\infty}}{-(s+a)} - \frac{e^0}{-(s+a)} = 0 + \frac{1}{(s+a)} = \frac{1}{(s+a)}_{//}
 \end{aligned}$$

$$3a. \int_{0^-}^{\infty} (e^{-a(t-T)} u(t-T)) e^{-st} dt$$

Applying the time shift property to result 3 tells us that the answer should be: $e^{-sT} \cdot \frac{1}{(s+a)}_{//}$

Let's do this from scratch:

Let $\tau = t - T \Rightarrow d\tau = dt$ (change of variable)

when $\begin{cases} t = 0^- , \tau = -T \\ t \rightarrow \infty , \tau \rightarrow \infty \end{cases}$ & $t = \tau + T$

Rewrite the integral:

$$\begin{aligned}
 &\int_{-T}^{\infty} e^{-a\tau} u(\tau) e^{-s(T+\tau)} d\tau \\
 &= e^{-sT} \int_{-T}^{\infty} \underbrace{u(\tau)}_{=1 \text{ when } \tau > 0} e^{-(s+a)\tau} d\tau
 \end{aligned}$$

(Take e^{-sT} outside since it is a constant)

$$= e^{-sT} \int_{0^+}^{\infty} 1 \cdot e^{-(s+a)\tau} d\tau$$

Lower limit changed to the region when unit step is 1.

(3)

$$= e^{-sT} \cdot \left[\frac{e^{-(s+a)t}}{-(s+a)} \right]_{0^+}^{\infty}$$

$$= e^{-sT} \cdot \left[\frac{e^{-\infty}}{-(s+a)} - \frac{e^0}{-(s+a)} \right] = e^{-sT} \left[0 + \frac{1}{(s+a)} \right]$$

$$= \frac{e^{-sT}}{(s+a)} //$$

$$4. \int_{0^-}^{\infty} t \underbrace{u(t)}_{=1 \text{ when } t > 0} e^{-st} dt = \int_{0^+}^{\infty} t e^{-st} dt$$

This requires integration by parts:

$$\int f(x) g(x) dx = f(x) \cdot G(x) - \int f'(x) G(x) dx$$

$$\text{where } G(x) = \int g(x) dx$$

Let's do the indefinite integration first:

$$\int t e^{-st} dt = \underset{\substack{\uparrow \\ f}}{t} \cdot \underset{\substack{\uparrow \\ G}}{\frac{e^{-st}}{(-s)}} - \int \underset{\substack{\uparrow \\ f'}}{1} \cdot \underset{\substack{\uparrow \\ G}}{\frac{e^{-st}}{(-s)}} dt$$

$$= -\frac{t}{s} e^{-st} + \frac{1}{s} \int e^{-st} dt$$

$$= -\frac{t}{s} e^{-st} + \frac{1}{s} \cdot \frac{e^{-st}}{(-s)}$$

$$= -\frac{t}{s} e^{-st} - \frac{1}{s^2} e^{-st}$$

(4)

Now let's apply the limits:

$$\left(-\frac{t}{s} e^{-st} - \frac{1}{s^2} e^{-st} \right) \Big|_{0^+}^{\infty}$$

$$= \left(\underset{\uparrow}{-0} - 0 \right) - \left(0 - \frac{1}{s^2} \cdot 1 \right) = \frac{1}{s^2} //$$

keep in mind that
 $\lim_{t \rightarrow \infty} (t e^{-st}) = 0$

Intuitively, the exponential goes down to 0 super fast compared to t 's increase.

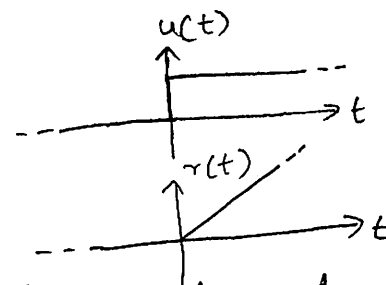
4a. $\int_{0^+}^{\infty} (t-T) u(t-T) dt$

Applying the time shift property to result 4 tells us that the answer should be $e^{-sT} \cdot \frac{1}{s^2} //$

Aside: we could have solved #4 by recognizing that

$t u(t) = r(t)$ ramp function,
 that can be obtained by integrating
 the step function

$$r(t) = \int_{0^+}^t u(z) dz$$



we can therefore apply the time integral property to $\mathcal{L}[u(t)] = \frac{1}{s}$, so we get $\frac{1}{s} \cdot \frac{1}{s} = \frac{1}{s^2}$ same answer //

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$$5. \mathcal{L}[t^2 u(t)] = \int_{0^-}^{\infty} t^2 u(t) e^{-st} dt$$

The easy way to do this is to recognize this as an integration of the ramp function.

$$\text{Remember that: } \int t \cdot dt = \frac{t^2}{2} \Rightarrow t^2 = 2 \cdot \int t dt$$

Using the time integral property:

$$\Rightarrow \mathcal{L}[t^2 u(t)] = 2 \cdot \frac{1}{s} \cdot \mathcal{L}[t u(t)] = \frac{2}{s} \cdot \frac{1}{s^2} = \frac{2}{s^3} //$$

Just for fun, let's do this from scratch:

$$\mathcal{L}[t^2 u(t)] = \int_{0^+}^{\infty} t^2 e^{-st} dt \quad (\text{because } u(t)=1, \text{ when } t>0)$$

Let's do the indefinite integral by parts first.

$$\int \underbrace{t^2}_f \underbrace{e^{-st}}_g dt = \underbrace{t^2}_f \cdot \underbrace{\left(\frac{e^{-st}}{-s}\right)}_{G_1} - \int \underbrace{2t}_{f'} \cdot \underbrace{\frac{e^{-st}}{(-s)}}_{G_1} dt$$

$$= -\frac{t^2 e^{-st}}{s} + \frac{2}{s} \underbrace{\int t \cdot e^{-st} dt}_{\text{Need to do integration by parts again!}}$$

$$= -\frac{t^2 e^{-st}}{s} + \frac{2}{s} \left[t \cdot \frac{e^{-st}}{(-s)} - \int 1 \cdot \frac{e^{-st}}{(-s)} dt \right]$$

$$= -\frac{t^2 e^{-st}}{s} - \frac{2t}{s^2} e^{-st} + \frac{2}{s^2} \int e^{-st} dt$$

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$$= -\frac{t^2 e^{-st}}{s} - \frac{2t}{s^2} e^{-st} + \frac{2}{s^2} \cdot \frac{e^{-st}}{(-s)}$$

$$= e^{-st} \left[-\frac{t^2}{s} - \frac{2t}{s^2} - \frac{2}{s^3} \right]$$

Now let's apply the limits:

$$e^{-st} \left[-\frac{t^2}{s} - \frac{2t}{s^2} - \frac{2}{s^3} \right] \bigg|_{t=0^+}^{t \rightarrow \infty}$$

keep in mind that the exponential falls off to 0 much faster than t or t^2 rise

$$= 0 - e^{-s \cdot 0} \left[0 - 0 - \frac{2}{s^3} \right]$$

$$= 2/s^3 //$$

$$6. \mathcal{L} [t e^{-at} u(t)] = \mathcal{L} [e^{-at} \cdot t u(t)]$$

The easiest way to do this problem is to apply the frequency shift property to $\mathcal{L} [t u(t)] = \frac{1}{s^2}$

what this property says is "replace every instance of s with $(s+a)$ ", so the answer

$$\text{is: } \frac{1}{(s+a)^2} //$$

$$7. \mathcal{L} [t^2 e^{-at} u(t)] = \mathcal{L} [\underbrace{e^{-at}} - \underbrace{t^2 u(t)}]$$

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$$\text{Now, } \mathcal{L} [t^2 u(t)] = \frac{2}{s^3}$$

we use the frequency shift property, replacing every instance of s with $(s+a)$

$$\Rightarrow \mathcal{L} [t^2 e^{-at} u(t)] = \frac{2}{(s+a)^3} //$$

$$8. \mathcal{L} [t^{n-1} e^{-at} u(t)]$$

This can be done easily using the frequency shift property if only we knew:

$$\mathcal{L} [t^{n-1} u(t)].$$

When you are faced with a problem like this, we should find a way to relate this formula to $\mathcal{L} [t^{n-2} u(t)]$. This will yield a recursion that leads all the way down to $\mathcal{L} [t^0 u(t)] = \frac{1}{s}$.

$$\begin{aligned} \mathcal{L} [t^{n-1} u(t)] &= \int_{0^-}^{\infty} t^{n-1} u(t) e^{-st} dt \\ &= \int_{0^+}^{\infty} t^{n-1} e^{-st} dt \end{aligned}$$

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Using integration by parts, we get:
(again, let's do the indefinite integral first)

$$\int \underbrace{t^{n-1}}_f \underbrace{e^{-st}}_g dt = \underbrace{t^{n-1}}_f \cdot \underbrace{\frac{e^{-st}}{(-s)}}_G - \int (n-1)t^{n-2} \cdot \frac{e^{-st}}{(-s)} dt$$

$$= -\frac{t^{n-1}}{s} \cdot e^{-st} + \frac{(n-1)}{s} \int t^{n-2} e^{-st} dt$$

Let's apply the limits:

$$-\frac{t^{n-1}}{s} \cdot e^{-st} \Big|_{0^+}^{\infty} + \frac{(n-1)}{s} \int_{0^+}^{\infty} t^{n-2} e^{-st} dt$$

$$= \left((-0) - 0 \right) + \frac{n-1}{s} \int_{0^+}^{\infty} t^{n-2} e^{-st} dt$$

Since t^{n-2} is a smooth function, this is the same as:

$$= \left(\frac{n-1}{s} \right) \int_{0^+}^{\infty} t^{n-2} e^{-st} dt = \left(\frac{n-1}{s} \right) \mathcal{L} [t^{n-2} u(t)] //$$

From this, we can guess that:

$$\mathcal{L} [t^{n-2} u(t)] = \left(\frac{n-2}{s} \right) \mathcal{L} [t^{n-3} u(t)]$$

This can go on all the way down to
 $\mathcal{L} [t^0 \cdot u(t)]$

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Therefore:

$$\mathcal{L}[t^{n-1} u(t)] = \left(\frac{n-1}{s}\right) \cdot \left(\frac{n-2}{s}\right) \cdot \dots \cdot \left(\frac{1}{s}\right)$$

$$= \frac{(n-1)!}{s^n} //$$

↖ factorial

Now, we can apply the frequency shift property (basically, replace all instances of s by $(s+a)$),

$$\Rightarrow \mathcal{L}[t^{n-1} e^{-at} u(t)] = \frac{(n-1)!}{(s+a)^n} //$$

Recognizing patterns:

Let us step back a bit and look at the following:

$$\mathcal{L}[u(t)] = \frac{1}{s}, \quad \mathcal{L}[t u(t)] = \frac{1}{s^2}, \quad \mathcal{L}[t^2 u(t)] = \frac{2}{s^3}$$

we can see that these formulas are all special cases of what we derived above, which can also be written as follows:

$$\mathcal{L}[t^n u(t)] = \frac{n!}{s^{n+1}}$$

$$\text{so } \mathcal{L}[t^2 u(t)] = \frac{2 \cdot 1}{s^3} //$$

9. $\mathcal{L} [\sin(\omega_0 t) u(t)] = \int_{0^-}^{\infty} \sin(\omega_0 t) u(t) e^{-st} dt$

The easiest way to do this problem is to use Euler's formula

$$\sin(\omega_0 t) = \frac{1}{2j} (e^{j\omega_0 t} - e^{-j\omega_0 t})$$

$$\Rightarrow \mathcal{L} [\sin(\omega_0 t) u(t)] = \frac{1}{2j} \mathcal{L} [e^{j\omega_0 t} u(t)] - \frac{1}{2j} \mathcal{L} [e^{-j\omega_0 t} u(t)]$$

$$= \frac{1}{2j} \left[\frac{1}{s - j\omega_0} - \frac{1}{s + j\omega_0} \right] = \frac{1}{2j} \cdot \frac{(s + j\omega_0) - (s - j\omega_0)}{(s^2 + \omega_0^2)} =$$

$$= \frac{\cancel{2j}\omega_0}{\cancel{2j}(s^2 + \omega_0^2)} = \frac{\omega_0}{s^2 + \omega_0^2} //$$

10. $\mathcal{L} [\sin(\omega_0 t + \theta) u(t)]$

$$= \mathcal{L} [(\sin(\omega_0 t) \cos \theta + \cos(\omega_0 t) \sin \theta) u(t)]$$

$$= \cos \theta \mathcal{L} [\sin(\omega_0 t) u(t)] + \sin \theta \mathcal{L} [\cos(\omega_0 t) u(t)]$$

$$= \cos \theta \cdot \left(\frac{\omega_0}{s^2 + \omega_0^2} \right) + \sin \theta \cdot \left(\frac{s}{s^2 + \omega_0^2} \right)$$

$$= \frac{(\omega_0 \cos \theta + s \sin \theta)}{(s^2 + \omega_0^2)} //$$

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$$11. \quad \mathcal{L}[\cos(\omega_0 t)u(t)]$$

$$= \mathcal{L}\left[\frac{1}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t})u(t)\right]$$

$$= \frac{1}{2}\mathcal{L}[e^{j\omega_0 t}u(t)] + \frac{1}{2}\mathcal{L}[e^{-j\omega_0 t}u(t)]$$

$$= \frac{1}{2} \cdot \frac{1}{(s - j\omega_0)} + \frac{1}{2} \cdot \frac{1}{(s + j\omega_0)}$$

$$= \frac{1}{2} \cdot \frac{s + j\omega_0 + s - j\omega_0}{s^2 + \omega_0^2} = \frac{s}{(s^2 + \omega_0^2)} //$$

$$12. \quad \mathcal{L}[\cos(\omega_0 t + \theta)u(t)]$$

$$= \mathcal{L}[(\cos(\omega_0 t)\cos\theta - \sin(\omega_0 t)\sin\theta)u(t)]$$

$$= \cos\theta \mathcal{L}[\cos(\omega_0 t)u(t)] - \sin\theta \mathcal{L}[\sin(\omega_0 t)u(t)]$$

$$= \cos\theta \cdot \frac{s}{(s^2 + \omega_0^2)} - \sin\theta \cdot \frac{\omega_0}{(s^2 + \omega_0^2)}$$

$$= \frac{s\cos(\theta) - \omega_0\sin(\theta)}{(s^2 + \omega_0^2)} //$$

13. $\mathcal{L}[e^{-at} \sin(\omega_0 t) u(t)]$

Now, $\mathcal{L}[\sin(\omega_0 t) u(t)] = \frac{\omega_0}{s^2 + \omega_0^2}$

Using Frequency shift:

$$\mathcal{L}[e^{-at} \sin(\omega_0 t) u(t)] = \frac{\omega_0}{(s+a)^2 + \omega_0^2} //$$

14. $\mathcal{L}[e^{-at} \cos(\omega_0 t) u(t)]$

Now, $\mathcal{L}[\cos(\omega_0 t) u(t)] = \frac{s}{s^2 + \omega_0^2}$

Using Frequency shift:

$$\mathcal{L}[e^{-at} \cos(\omega_0 t) u(t)] = \frac{(s+a)}{(s+a)^2 + \omega_0^2} //$$

15. $\mathcal{L}[2e^{-at} \cos(bt-\theta) u(t)]$

First, let's figure out $\mathcal{L}[2\cos(bt-\theta) u(t)]$

$$= \mathcal{L}[(e^{j(bt-\theta)} + e^{-j(bt-\theta)}) u(t)]$$

$$= e^{-j\theta} \mathcal{L}[e^{jbt} u(t)] + e^{+j\theta} \mathcal{L}[e^{-jbt} u(t)]$$

$$= \frac{e^{-j\theta}}{s - jb} + \frac{e^{+j\theta}}{s + jb}$$

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Now let us apply the frequency shift by replacing s by $(s+a)$ throughout.

$$\Rightarrow \mathcal{L} [2e^{-at} \cos(bt-\theta) u(t)]$$

$$= \frac{e^{-j\theta}}{s+a-jb} + \frac{e^{j\theta}}{s+a+jb} //$$

15 a. $\mathcal{L} [e^{-at} \cos(bt-\theta) u(t)]$

This is the same problem as #15, but we seek a different format:

$$\mathcal{L} [\cos(bt-\theta) u(t)]$$

$$= \mathcal{L} [(\cos(bt)\cos\theta + \sin(bt)\sin\theta) u(t)]$$

$$= \cos\theta \mathcal{L} [\cos(bt) u(t)] + \sin\theta \mathcal{L} [\sin(bt) u(t)]$$

$$= \cos\theta \cdot \frac{s}{s^2+b^2} + \sin\theta \cdot \frac{b}{s^2+b^2}$$

Now apply frequency shift, replacing s by $(s+a)$

$$\mathcal{L} [e^{-at} \cos(bt-\theta) u(t)]$$

$$= \frac{(s+a)\cos\theta + b\sin\theta}{(s+a)^2 + b^2} //$$

This looks formidable at first sight, but we can recognize the components A, B & C corresponding to #8, and #15

we know that $\mathcal{L}[t^{n-1} u(t)] = \frac{sn}{(n-1)!}$ (page 9)

At this point, Euler's formula is our best friend since it can convert the cosine into exponentials that can be manipulated easily:

$$2 \cos(bt - \theta) = (e^{j(bt - \theta)} + e^{-j(bt - \theta)})$$

$$= e^{-at} \cos(bt - \theta) + e^{-at - j(bt - \theta)} + e^{-at + j(bt - \theta)} + e^{-at - (a + j)b t}$$

$$\Rightarrow \mathcal{L}\left[\frac{2t^{n-1}}{2} e^{-at} \cos(bt - \theta) u(t) \right]$$

$$= \mathcal{L}\left[\left(\frac{t^{n-1}}{(n-1)!} \cdot e^{-j\theta} \cdot e^{-j\omega t} - (a + j)b t \cdot \frac{t^{n-1}}{(n-1)!} \cdot e^{-j\omega t} \right) \cdot e^{-at} \right]$$

$$= e^{-j\theta} \mathcal{L}\left[\frac{t^{n-1}}{(n-1)!} \cdot e^{-j\omega t} \right] + e^{-j\theta} \mathcal{L}\left[\frac{t^{n-1}}{(n-1)!} \cdot e^{-j\omega t} \cdot (-a - jb) \right]$$

$$= \frac{e^{-j\theta}}{(n-1)!} \cdot \mathcal{L}\left[\frac{t^{n-1}}{(n-1)!} \cdot e^{-j\omega t} \right] + \frac{e^{-j\theta}}{(n-1)!} \cdot \mathcal{L}\left[\frac{t^{n-1}}{(n-1)!} \cdot e^{-j\omega t} \cdot (-a - jb) \right]$$

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$$= \frac{e^{-j\theta}}{\cancel{(n-1)!}} \cdot \frac{\cancel{(n-1)!}}{(s+a-jb)^n} + \frac{e^{j\theta}}{\cancel{(n-1)!}} \cdot \frac{\cancel{(n-1)!}}{(s+a+jb)^n}$$

$$= \frac{e^{-j\theta}}{(s+a-jb)^n} + \frac{e^{j\theta}}{(s+a+jb)^n} //$$

☺